UNIT 8 — METALS AND NON-METALS

PART I — COMPLETE REVISION NOTES

1. ELEMENTS AND THEIR CLASSIFICATION

An element is a pure substance made up of only one kind of atom. Elements are broadly classified as metals, non-metals and metalloids.

• Metals: commonly lustrous, malleable, ductile and good conductors of heat and electricity. Examples: iron, copper, aluminium, gold, silver, sodium and lead.

• Non-metals: commonly dull, brittle when solid, non-malleable and poor conductors. Examples: oxygen, hydrogen, sulphur, carbon, nitrogen and bromine.

• Metalloids: show some properties of both metals and non-metals. Examples: boron, silicon, germanium, arsenic and antimony. Many are semiconductors.

2. DIFFERENCES BETWEEN METALS AND NON-METALS

Property	Metals (general trend)	Non-metals (general trend)
Lustre	Usually shiny	Usually dull
Density	Often high	Often low
Malleability	Can be beaten into sheets	Usually brittle when solid
Ductility	Can be drawn into wires	Usually non-ductile
Conductivity	Usually good conductors	Usually poor conductors
Melting point	Often high	Often low

Important exceptions: Mercury is a liquid metal at room temperature. Bromine is a liquid non-metal. Graphite (a form of carbon) conducts electricity. Some metals, such as sodium and potassium, are soft and have relatively low melting points. These are general properties, not absolute rules.

3. METALLOIDS

• They lie near the boundary between metals and non-metals in the periodic table.

• They may show metallic as well as non-metallic characteristics.

• Silicon and germanium are important semiconductor materials used in electronic devices.

4. SYMBOLS OF ELEMENTS

A chemical symbol is the shortened representation of an element. It usually contains one or two letters. The first letter is capital and the second, if present, is lowercase. J. J. Berzelius introduced the modern letter-symbol system.

Element	Symbol	Element	Symbol
Hydrogen	H	Helium	He
Lithium	Li	Carbon	C
Nitrogen	N	Oxygen	O
Fluorine	F	Neon	Ne
Magnesium	Mg	Aluminium	Al
Silicon	Si	Phosphorus	P
Sulphur	S	Chlorine	Cl
Calcium	Ca	Iron	Fe
Copper	Cu	Silver	Ag

Symbols derived from Latin names

Element	Symbol	Latin name
Sodium	Na	Natrium
Potassium	K	Kalium
Iron	Fe	Ferrum
Copper	Cu	Cuprum
Silver	Ag	Argentum
Gold	Au	Aurum
Mercury	Hg	Hydrargyrum
Lead	Pb	Plumbum
Tin	Sn	Stannum

5. ALLOTROPY

Allotropy is the property of an element to exist in two or more different forms in the same physical state. These forms have different physical properties but similar chemical properties.

• Carbon: diamond, graphite and fullerene are crystalline allotropes. Coal, coke, charcoal and lampblack are amorphous forms of carbon.

• Phosphorus: white, red and black phosphorus.

• Sulphur: crystalline forms include rhombic (alpha) and monoclinic (beta) sulphur. Amorphous forms include plastic, colloidal and milk of sulphur.

6. DIAMOND AND GRAPHITE

Property	Diamond	Graphite
Hardness	Extremely hard; hardest natural material	Soft and slippery
Appearance	Pure diamond is colourless and transparent	Usually black/grey and opaque
Electrical conductivity	Poor conductor	Good conductor
Structure	Each carbon bonded to four other carbon atoms in a 3-D network	Each carbon bonded to three others in hexagonal layers
Density (approx.)	3.5 g/cm³	2.25 g/cm³
Uses	Cutting tools, jewellery	Pencil 'lead', electrodes, lubricants

7. FULLERENES

• Fullerenes are crystalline allotropes of carbon made of hollow cage-like or tube-like structures.

• A well-known fullerene, buckminsterfullerene (C■■), contains 60 carbon atoms arranged in a cage-like structure.

• Potential and research applications include materials science, electronics, catalysis and drug-delivery research.

8. AMORPHOUS FORMS OF CARBON

Form	Preparation / property	Uses
Wood charcoal	Produced by heating wood with limited or no air; porous and adsorbent	Fuel; adsorption of some gases and impurities
Animal / bone charcoal	Obtained by heating bones in absence of air; can adsorb coloured substances	Decolourising and purification processes
Coke	Obtained by destructive distillation of coal; hard, greyish-black solid	Fuel and reducing agent in extraction of iron; electrodes
Lampblack	Produced by incomplete burning of carbon-rich materials	Printing ink, paints, polish and rubber products

9. COAL AND CARBONISATION

• Coal is a fossil fuel formed from buried plant material over millions of years under heat and pressure in the absence of air.

• Carbonisation is the slow conversion of buried plant material into coal.

• Coal is a complex mixture containing mainly carbon along with compounds of hydrogen, oxygen, nitrogen and sulphur.

Coal type	Approx. carbon content
Peat	60%
Lignite (soft / brown coal)	70%
Bituminous (household coal)	80%
Anthracite (hard coal)	90%

Uses of coal

• Used as a domestic and industrial fuel.

• Used to produce fuel gases and as a source of coke and other industrial products.

10. PETROLEUM

• Petroleum (crude oil), often called black gold, is a naturally occurring dark, viscous mixture of hydrocarbons.

• It formed from the remains of ancient plants and animals buried under sediments and transformed over millions of years.

• Crude petroleum is a mixture, not a single pure compound. It is generally lighter than water and insoluble in it.

• Refining is the separation of crude petroleum into useful fractions by fractional distillation, based on differences in boiling ranges.

Important petroleum fractions include petroleum gas, petrol (gasoline), kerosene, diesel, lubricating oil, paraffin wax and bitumen. Exact fraction names and ranges depend on the refining process.

11. FUELS AND THEIR TYPES

A fuel is a substance that releases useful heat energy when it burns in oxygen.

Type	Examples	General features
Solid	Wood, charcoal, coal, coke, paraffin wax	Often need more storage space; may leave ash
Liquid	Petrol, kerosene, diesel, fuel oil	Convenient to transport and store; generally leave little solid residue
Gaseous	Natural gas, LPG, biogas, producer gas, water gas	Can burn cleanly when properly supplied with air; easy to control

12. IMPORTANT FUEL GASES

Gas	Main composition / preparation	Key point
Natural gas	Mainly methane (CH■)	Fossil fuel; widely used as a fuel
Producer gas	Mainly carbon monoxide (CO) and nitrogen (N■); made by passing air over hot coke	Nitrogen dilutes the fuel gas, lowering its heating value
Water gas	Mainly carbon monoxide (CO) and hydrogen (H■); steam passed over red-hot coke	Both major components are combustible
Coal gas	Mixture including hydrogen, methane and carbon monoxide; produced during destructive distillation of coal	Used as a fuel in some industrial contexts
Biogas / gobar gas	Mainly methane with carbon dioxide and traces of other gases; formed by anaerobic decomposition	Renewable fuel made from organic waste
LPG	Mainly propane and butane; stored as a liquid under pressure	Odourant such as ethyl mercaptan is added to help detect leaks

Advantages of LPG

• High calorific value and convenient storage in cylinders.

• Burns with a relatively clean flame when combustion is complete.

• Ethyl mercaptan gives the otherwise nearly odourless gas a detectable smell in case of leakage.

13. EXAM-IMPORTANT FACTS

• Mercury is a liquid metal; bromine is a liquid non-metal at ordinary room temperature.

• Graphite conducts electricity, whereas diamond is a poor electrical conductor.

• Diamond is extremely hard; graphite is soft because its layers can slide over one another.

• Peat has the lowest carbon content and anthracite the highest among the four coal types listed.

• Producer gas contains nitrogen; water gas mainly contains hydrogen and carbon monoxide.

• LPG is mainly propane and butane; commercial composition can vary.

• Petroleum is a mixture of hydrocarbons; refining separates it into fractions.

• Symbols from Latin names: Na, K, Fe, Cu, Ag, Au, Hg, Pb and Sn.

PART II — 90 MULTIPLE-CHOICE QUESTIONS

Choose the one best answer for each question. Each question carries 1 mark. Suggested time: 75 minutes.

1. An element is a pure substance made up of:

(A) One kind of atom (B) Two kinds of molecules
(C) Different mixtures (D) Only ions

2. Which group contains only metals?

(A) Oxygen, sulphur, carbon (B) Iron, copper, silver
(C) Bromine, chlorine, nitrogen (D) Silicon, arsenic, boron

3. Which of the following is a metalloid?

(A) Sodium (B) Sulphur
(C) Silicon (D) Gold

4. Which property is commonly shown by metals?

(A) Malleability (B) Brittleness in solid state
(C) Poor conductivity (D) Dullness

5. Which of the following is a non-metal?

(A) Copper (B) Aluminium
(C) Lead (D) Sulphur

6. A metal that is liquid at ordinary room temperature is:

(A) Mercury (B) Iron
(C) Sodium (D) Copper

7. A non-metal that is liquid at ordinary room temperature is:

(A) Carbon (B) Bromine
(C) Sulphur (D) Phosphorus

8. Which non-metal is a good conductor of electricity?

(A) Sulphur (B) Oxygen
(C) Graphite (D) Phosphorus

9. Metalloids are called semiconductors because they:

(A) Never conduct electricity (B) Conduct only when molten
(C) Conduct better than all metals (D) Have conductivity between conductors and insulators under suitable conditions

10. Which is NOT a typical property of metals?

(A) Ductility (B) Malleability
(C) Good electrical conductivity (D) Poor electrical conductivity

11. The ability of a metal to be drawn into wires is called:

(A) Ductility (B) Malleability
(C) Sonority (D) Lustre

12. The ability of a metal to be beaten into thin sheets is:

(A) Ductility (B) Malleability
(C) Elasticity (D) Brittleness

13. Which is the correct symbol for sodium?

(A) S (B) So
(C) Na (D) N

14. The symbol K is derived from the Latin name:

(A) Kalium (B) Krypton
(C) Kalsium (D) Karbon

15. The Latin name Ferrum gives the symbol for:

(A) Fluorine (B) Iron
(C) Silver (D) Tin

16. Which pair is correctly matched?

(A) Gold–Ag (B) Silver–Au
(C) Copper–Cu (D) Lead–Sn

17. The symbol Hg represents:

(A) Hydrogen (B) Helium
(C) Gold (D) Mercury

18. Which element has the symbol Pb?

(A) Platinum (B) Lead
(C) Phosphorus (D) Potassium

19. Which is the correct way to write a two-letter chemical symbol?

(A) Both letters lowercase (B) Both letters uppercase
(C) First uppercase, second lowercase (D) First lowercase, second uppercase

20. Who introduced the modern system of representing elements by letters?

(A) J. J. Berzelius (B) John Dalton
(C) Isaac Newton (D) Dmitri Mendeleev

21. Allotropy is the ability of an element to:

(A) Form only compounds (B) Exist in different forms in the same physical state
(C) Dissolve in water (D) Change into another element

22. Which set contains allotropes of carbon?

(A) Diamond, graphite, fullerene (B) Coal, iron, coke
(C) Red, white, black phosphorus (D) Rhombic, monoclinic, plastic sulphur

23. Which is NOT an allotrope of phosphorus?

(A) White phosphorus (B) Red phosphorus
(C) Black phosphorus (D) Graphite

24. Which are crystalline forms of sulphur?

(A) White and red (B) Diamond and graphite
(C) Rhombic and monoclinic (D) Coke and charcoal

25. Which is an amorphous form of sulphur?

(A) Diamond (B) Plastic sulphur
(C) Graphite (D) Fullerene

26. In diamond, each carbon atom is bonded to how many other carbon atoms?

(A) Two (B) Three
(C) Four (D) Six

27. In graphite, each carbon atom is bonded to:

(A) Three other carbon atoms (B) Four other carbon atoms
(C) Two other carbon atoms (D) One other carbon atom

28. Graphite conducts electricity mainly because it has:

(A) No carbon atoms (B) Free-moving electrons within its structure
(C) Only ionic bonds (D) A liquid structure

29. Which form of carbon is the hardest natural substance?

(A) Graphite (B) Coke
(C) Charcoal (D) Diamond

30. Which allotrope of carbon is used in pencil leads?

(A) Diamond (B) Graphite
(C) Fullerene (D) Lampblack

31. Pure diamond is generally:

(A) Black and opaque (B) Soft and greasy
(C) Colourless and transparent (D) A good electrical conductor

32. The layers in graphite can slide over one another, making it:

(A) Soft and slippery (B) Extremely hard
(C) A liquid (D) Transparent

33. Buckminsterfullerene commonly contains:

(A) 12 carbon atoms (B) 24 carbon atoms
(C) 60 carbon atoms (D) 100 carbon atoms

34. Which is an amorphous form of carbon?

(A) Diamond (B) Graphite
(C) Fullerene (D) Coke

35. Wood charcoal is generally prepared by heating wood:

(A) In excess oxygen (B) With limited or no air
(C) In water (D) With hydrogen

36. A useful property of wood charcoal is its:

(A) High transparency (B) Magnetic nature
(C) Adsorbing ability (D) High electrical conductivity

37. Bone charcoal is obtained from:

(A) Bones heated in absence of air (B) Petroleum distillation
(C) Melting graphite (D) Burning sulphur

38. Coke is prepared by destructive distillation of:

(A) Petroleum (B) Wood
(C) Natural gas (D) Coal

39. Coke is used in the extraction of iron because it acts as:

(A) An indicator (B) A reducing agent and fuel
(C) A lubricant only (D) A coolant

40. Lampblack is commonly used in the manufacture of:

(A) Glass lenses (B) Fertilisers
(C) Printing ink (D) Cement only

41. Coal was formed mainly from ancient:

(A) Marine salts (B) Plant material
(C) Pure metals (D) Volcanic glass

42. The slow conversion of buried plant material into coal is called:

(A) Carbonisation (B) Refining
(C) Sublimation (D) Neutralisation

43. Which type of coal has the highest carbon content in the given textbook table?

(A) Peat (B) Lignite
(C) Bituminous (D) Anthracite

44. Which type of coal has approximately 60% carbon in the table?

(A) Anthracite (B) Bituminous
(C) Peat (D) Lignite

45. The carbon content of lignite in the textbook table is approximately:

(A) 60% (B) 70%
(C) 80% (D) 90%

46. Which is often called black gold?

(A) Coal (B) Charcoal
(C) Coke (D) Petroleum

47. Crude petroleum is:

(A) A pure element (B) A mixture of hydrocarbons
(C) A single compound (D) A metal ore

48. Petroleum is generally:

(A) Lighter than water and insoluble in it (B) Heavier than water and soluble in it
(C) Completely miscible with water (D) A solid at room temperature

49. The separation of crude petroleum into useful fractions is called:

(A) Carbonisation (B) Crystallisation
(C) Refining (D) Filtration

50. Petroleum fractions are separated mainly by differences in:

(A) Colour (B) Boiling ranges
(C) Magnetism (D) Particle shape

51. Which of the following is a petroleum fraction?

(A) Kerosene (B) Oxygen
(C) Sulphur (D) Graphite

52. Which is a solid fuel?

(A) Petrol (B) LPG
(C) Diesel (D) Coal

53. Which is a liquid fuel?

(A) Wood (B) Kerosene
(C) Coke (D) Charcoal

54. Which is a gaseous fuel?

(A) Petrol (B) Coal
(C) Natural gas (D) Coke

55. Natural gas consists mainly of:

(A) Methane (B) Oxygen
(C) Nitrogen (D) Hydrogen chloride

56. The main combustible components of water gas are:

(A) Nitrogen and oxygen (B) Hydrogen and carbon monoxide
(C) Carbon dioxide and nitrogen (D) Methane and oxygen

57. Producer gas mainly contains:

(A) Hydrogen and oxygen (B) Methane and carbon dioxide
(C) Nitrogen and carbon monoxide (D) Only carbon dioxide

58. Producer gas has a lower heating value than water gas mainly because it contains:

(A) Hydrogen (B) Carbon monoxide
(C) Methane (D) Nitrogen

59. Water gas is prepared by passing which substance over red-hot coke?

(A) Steam (B) Air only
(C) Oxygen only (D) Petrol vapour

60. Coal gas is obtained during the destructive distillation of:

(A) Wood (B) Coal
(C) Petroleum (D) Natural gas

61. Biogas is produced by:

(A) Burning coal in air (B) Distilling crude oil
(C) Anaerobic decomposition of organic matter (D) Melting limestone

62. Gobar gas contains mainly:

(A) Oxygen (B) Nitrogen
(C) Hydrogen (D) Methane

63. LPG mainly contains:

(A) Propane and butane (B) Hydrogen and oxygen
(C) Nitrogen and carbon dioxide (D) Methane and chlorine

64. LPG is stored in cylinders as a liquid because it is:

(A) Cooled below absolute zero (B) Compressed under pressure
(C) Mixed with water (D) Converted into a solid

65. Why is ethyl mercaptan added to LPG?

(A) To increase its colour (B) To reduce its pressure
(C) To help detect gas leakage by smell (D) To make it non-flammable

66. Which is an advantage of LPG when properly burned?

(A) Produces large amounts of ash (B) Always produces smoke
(C) Cannot be stored (D) Burns with a relatively clean flame

67. Which fuel is renewable when produced from organic waste?

(A) Biogas (B) Coal
(C) Petroleum (D) Coke

68. Which pair contains only fossil fuels?

(A) Wood and biogas (B) Coal and petroleum
(C) Biogas and wood (D) Hydrogen and biogas

69. Which statement about fuels is correct?

(A) A fuel releases useful heat energy on burning (B) All fuels are solids
(C) All fuels leave ash (D) All fuels contain only carbon

70. Which of these is NOT a fuel gas?

(A) Natural gas (B) Producer gas
(C) Water gas (D) Graphite

71. Which material is commonly used as an adsorbent in gas masks?

(A) Wood charcoal / activated charcoal (B) Pure diamond
(C) Sodium metal (D) Sulphur crystals

72. Which element is a metalloid used widely in semiconductor devices?

(A) Gold (B) Silicon
(C) Sulphur (D) Bromine

73. Which set contains only symbols derived from Latin names?

(A) H, O, N (B) C, S, P
(C) Na, K, Fe (D) He, Ne, Ar

74. Which is the correct symbol for silver?

(A) Si (B) S
(C) Sr (D) Ag

75. Which element is represented by Sn?

(A) Tin (B) Sodium
(C) Sulphur (D) Silicon

76. Which element is represented by Au?

(A) Aluminium (B) Gold
(C) Argon (D) Silver

77. Which property explains why copper is used for electrical wiring?

(A) Dullness (B) Brittleness
(C) Good electrical conductivity and ductility (D) Low density only

78. Which object is most likely made from a malleable metal?

(A) A thin aluminium foil sheet (B) A sulphur crystal
(C) A charcoal lump (D) A graphite powder cake

79. A solid material is shiny, can be drawn into wires and conducts electricity well. It is most likely a:

(A) Non-metal (B) Metal
(C) Gas (D) Mixture of gases

80. A substance is brittle, dull and a poor conductor. It is most likely a:

(A) Metal (B) Metalloid in every case
(C) Non-metal (D) Liquid metal

81. Which statement about metalloids is correct?

(A) They have only metallic properties (B) They have only non-metallic properties
(C) They are all liquids (D) They show some properties of metals and non-metals

82. Which pair is correctly matched?

(A) Mercury–liquid metal (B) Bromine–liquid metal
(C) Graphite–electrical insulator (D) Diamond–soft lubricant

83. Which of the following is NOT a form of carbon?

(A) Fullerene (B) Graphite
(C) Red phosphorus (D) Diamond

84. A student wants a substance for a pencil core. Which should be selected?

(A) Diamond (B) Graphite
(C) Coke (D) Bone charcoal

85. Which coal type is called soft coal in the given table?

(A) Lignite (B) Anthracite
(C) Peat (D) Bituminous

86. Which of the following is a correct sequence of increasing carbon content in the table?

(A) Anthracite, bituminous, lignite, peat (B) Lignite, peat, anthracite, bituminous
(C) Bituminous, lignite, peat, anthracite (D) Peat, lignite, bituminous, anthracite

87. Which is the best explanation for why crude petroleum is not a pure substance?

(A) It is always solid (B) It contains only one kind of atom
(C) It contains many different hydrocarbons and other substances (D) It is a single element

88. Which fuel gas is mainly a mixture of hydrogen and carbon monoxide?

(A) Producer gas (B) Water gas
(C) Natural gas (D) LPG

89. Which fuel gas contains nitrogen as a major component along with carbon monoxide?

(A) Producer gas (B) Water gas
(C) Biogas (D) LPG

90. Which one is a useful application of coke?

(A) Making litmus (B) Producing milk of sulphur
(C) Making diamond jewellery (D) Reducing iron ore in a blast furnace

ANSWER KEY — 90 MCQs

1. A 2. B 3. C 4. A 5. D 6. A 7. B 8. C 9. D 10. D

11. A 12. B 13. C 14. A 15. B 16. C 17. D 18. B 19. C 20. A

21. B 22. A 23. D 24. C 25. B 26. C 27. A 28. B 29. D 30. B

31. C 32. A 33. C 34. D 35. B 36. C 37. A 38. D 39. B 40. C

41. B 42. A 43. D 44. C 45. B 46. D 47. B 48. A 49. C 50. B

51. A 52. D 53. B 54. C 55. A 56. B 57. C 58. D 59. A 60. B

61. C 62. D 63. A 64. B 65. C 66. D 67. A 68. B 69. A 70. D

71. A 72. B 73. C 74. D 75. A 76. B 77. C 78. A 79. B 80. C

81. D 82. A 83. C 84. B 85. A 86. D 87. C 88. B 89. A 90. D

ALL THE BEST